C2W Hybrid Bonding Interconnect Technology for Higher Density and Better Thermal Dissipation of High Bandwidth Memory

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Abstract - Cu/Oxide Hybrid Bonding Interconnect (HBI) technology has been used in the mass production of CMOS Image Sensors and advanced packages for fine pitch heterogeneous interconnection. Furthermore, HBI can be implemented for high density products demanding good heat dissipation due to HBI's structure lacking micro bumps and polymer underfill. Thus, HBI is widely studied in the field of semiconductor assembly. Recently, we have demonstrated HBI technology in chip-to-wafer (C2W) multi stack High Bandwidth Memory (HBM) real devices and obtained promising results. We developed wafer supporting system materials to maintain Cu pad clean and focused on optimizing plasma activation conditions for high bond strength and minimizing change in Cu recess level for stable Cu/Cu joints even in low temperature annealing. Even a single void induced by various particles can cause electrical failures and reliability problems. In C2W, particles can be introduced by various processes and environments prior to bonding, unlike in wafer on wafer (WOW). Hence, an all-in-one stacking equipment with a pre-treatment function was developed to prevent such particles. Through several core technological improvements, we were able to obtain better cumulative yields than anticipated from HBM 8Hi real devices. Although further researches on C2W HBI technology are needed, we have obtained highly reliable HBM through HBI. HBI implemented HBM showed good electrical functions similar to that of conventional HBM, but had a thermal resistance at less than 20% of that of conventional HBM. In addition, we confirmed that HBI can be a powerful solution for 16hi HBM since the stack height can be reduced by more than 15% of conventional HBM. Since HBI has many advantages, we expect it to be widely used in 3D/2.5D packages in the near future.

Keywords – Hybrid bonding, Chip to Wafer Bonding, High Bandwidth Memory, Thermal Resistivity

I. INTRODUCTION

The importance of HBM is rapidly growing alongside the recent development of markets demanding high performance computing, such as Artificial Intelligence, Internet of Things and machine learning. Until now, HBM has been connected through micro solder bumps between chips, but it is required to reduce pitch in order to cope with higher bandwidth. And chip thickness should be thin to make high capacity memory. HBI could be potential candidate for such next generation HBM, as Cu pads and the thin polished passivation layer of HBI can replace the micro solder bumps and polymer underfill, scaling down the bump pitch. Compared to the conventional solder bump structure, HBI can reduce the stacked thickness by 10~15% as shown in Fig.1 or allow for an increase in chip thickness to enable a more stable wafer level processing. However, HBI has yet to overcome problems for adoption to HBM, as HBM requires a high (12+) level of C2W stacking. Unlike wafer to wafer bonding process, chip to wafer HBI should has backside pads for multi-chip interconnection. So HBI should focus on Cu pad management because Cu pad can be easily deteriorated during many thermal processes after front side CMP. Emphasis should also be given to what is trivial at low stacks, but is consequential at high stacks, such as stack voids by particles and cumulative erosion. HBI for HBM is still at the development stage, and may be immature for mass production. But the images and data in this paper are from real HBM devices, not test vehicles, providing concrete evidence that the mass production of HBM using HBI is highly plausible. We hope that this paper could provide insights to diagnose the possibility of HBM in HBI.

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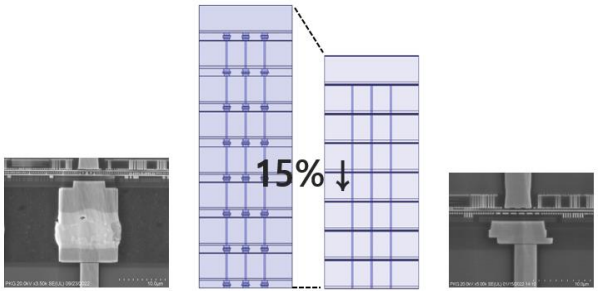


Fig.1 Structure comparison between Solder bump and HBI (in case of same chip thickness of 58um)

II. RESULTS

A. Material Development and Process Opimization for Temporary Bonding Process

For multi-stack chips like HBM, temporary bonding and debonding (TBDB) process is required because bumps or pads must be formed on the front and back side of each layer. There are several methods and materials used for conventional solder bump structure. However, since Cu pads in HBI can be easily oxidized, materials and process conditions need be carefully implemented.

A cure step during temporary bonding process should be required for stable back side processing. Fig.2 shows Cu pad quality after wafer de-bonding process, at different cure conditions. After long annealing time and high temperature, the Cu pad deformed to a protruding topology, which led to deteriorated inter-passivation bond strength. A short annealing time at mid-temperature led to Cu pad oxidation and thus discoloration. As discolored Cu pads form poor Cu/Cu joints, adhesives used to temporarily bond the Cu pad interfaces should be chosen carefully. We established main parameters for optimal Cu pads, such as annealing times, curing temperatures, Q-time between front side CMP and debonding as well as additives for Cu oxidation inhibition.

Materials	A	B	B'	C	C
Cure Conditions	5min@210'C	2hrs@190'C	10min@200'C	10min@200'C	2hrs@190'C
Cu Recess Variance	None	+60nm	None	None	+10nm
Cu Oxide					

Fig. 2 Optical image after de-bonding by TBDB conditions

B. Plasma Treatment Process Optimization for Reliability and Long Q-time at stacking process

Plasma treatment is a main process that makes oxide bonding at room temperature with CMP in HBI. In the early development stage, oxide bonding with low bonding strength could not endure mechanical stress under reliability conditions especially in moisture sensitivity level (MSL) test and highly accelerated temperature and humidity stress test (HAST). So continuous processing with 2 types of gases which increases the OH^- density at the interface was applied without roughness increase of passivation layer. However, edge voids were detected in the stacked chips when the Q-time between the plasma process and the stack process exceeded 4hrs. Thus, plasma treatment parameters were adjusted such that the samples were void and delamination free even at a time window of 24hrs, as shown in Fig.4. Above bonding strength is measured by the Maszara test in wafer to wafer base.

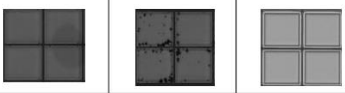
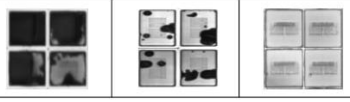
Plasma Conditions		A	B	C
Bond Strength		1.0J	2.5J	2.0J
SAM	End of Line @ TV			
	After Reliability @ Real device			

Fig. 4 SAT Results of End of line and reliability test by plasma conditions

C. Cu Recess Management for Cu/Cu Joint at Low Temp. Annealing

Annealing conditions for HBM in HBI should be limited to below 200℃ to prevent the over-growth of base wafer solder intermetallic compounds, and refresh degradation of DRAM. Data from Scanning Probe Microscopes showed that Cu pads with 10um diameter expanded vertically by approximately 10nm at 200℃, as shown in Fig.5. So Cu recess should be carefully controlled to prevent joint failure.

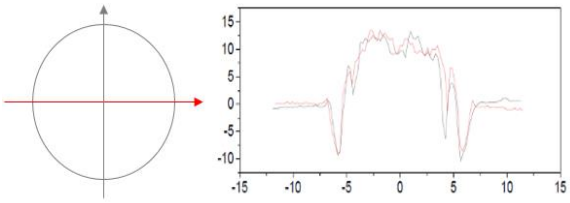


Fig.5 Measurement of Cu Pop-up at 200'C

So Cu recess should be carefully controlled to prevent joint failure. Joints failed at Cu recess level of -4nm during /after the CMP process. Measurements of Cu recess for each step between front side CMP and stacking confirmed that Cu recess grew rapidly during the mega-sonic cleaning process, a process performed to remove particles. Fig.6 also shows a positive correlation between dishing and Cu recess. CO_2 in De-ionized water, used to prevent electro-static discharge contributed to additional recess formation and Fig.6 also shows that the larger dishing is, the more recesses is occurred.

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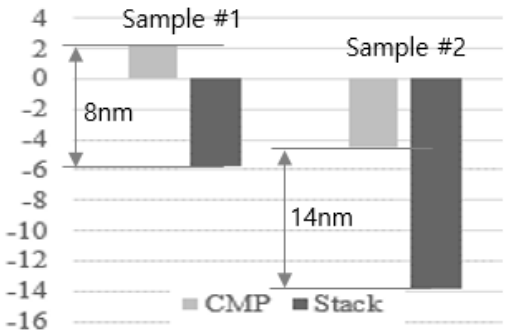

Fig.6 Cu Recess results according to process step

So de-ionized water with reduced CO_2 concentration was used, and the cleaning time was shortened to minimize the Cu recess. After optimization of the cleaning and CMP conditions, Cu/Cu joints did not fail even at low temperature annealing. It is expected to be additionally required of special cleaning method not affecting Cu recess but removing particles easily.

D. Minimization of Stack Voids

Even a single void can cause electrical failure and reliability problem. In the HBI stack process, void can be generated by various reasons, including outgassing of passivation film during the annealing process, and uneven flatness during stack process. Additional processes such as pre-bake used in the c-mos image sensor field to prevent outgassing are difficult to implement, and uneven flatness is exacerbated, due to the multi-layer stacked structure of the HBM package. As an alternative, we have developed a low-outgassing passivation film, redesigned the Cu pad, and optimized CMP conditions such as the slurry and process parameters.

However, the biggest cause of stack voids is air trap formed during the stack process, originating from different kinds of particles and sequential bonding failure form the center. In the early stage of development, a suitable convex tool that could initiate the center of chip could easily remove the void trap. However, in real chip with large warpage and uneven flatness, having perfect flatness, stack void could not be removed by the hard type convex tool. After experiments to optimize main factors such as tool design and material properties, the air trap induced stack voids could be removed as shown in Fig.7. In addition, adjustments to bond parameters including force, time and speed were required for the real chips to be stack void free.

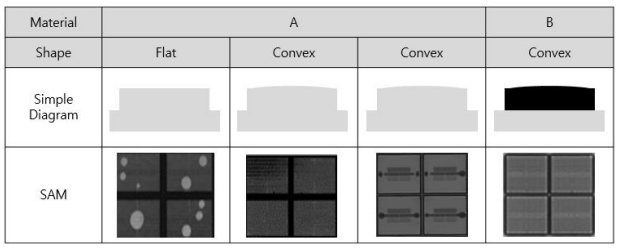

Fig.7 Air trap according to bond tool

Unlike chips with bumps or die attach films, HBM with HBI has no gaps, and particles lead to voids. Scanning Acoustic Tomography shows that thin and soft particles may not cause voids, but on signal pads, particles results in voids that cause yield losses, as shown in Fig.8.

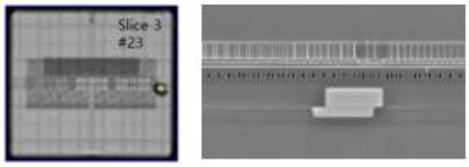

Fig.8 SAM with failure map vs. failure analysis results

In case of HBI process, many kinds of foreign materials were occurred by dicing process including stealth dicing and expand-cut. Also, environmental particles could be originated by storage and even transportations as shown in Fig.9.

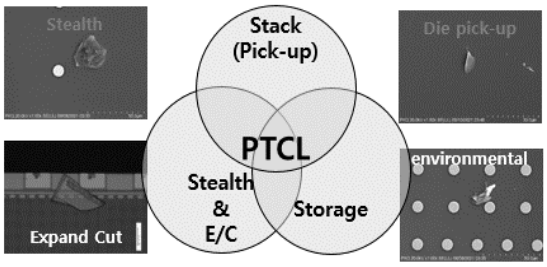

Fig. 9 Major foreign materials

Dicing process – The front side of wafer should be contacted on chuck table during stealth dicing and silicon debris cannot be avoidable due to expand-cut process. Most of such foreign materials originated by dicing process could be easily removed by applying protection layer during dicing process and high pressure cleaning after dicing process. However, when the residue of the protection layer remains after processes, it leads to yield loss regardless of residue size. Therefore, we are still working on improving the solubility of protection layer and finding new cleaning methods.

Stacking process – Chip to wafer stacking process involves die pick-up, chip transfer and bonding, which requires repeated transport of individual dies, and a long processing time unlike wafer to wafer bonding process. This increases the risk of contamination from continuous movement and prolonged exposure time. In particular, the pick-up area is much more contaminated than other areas, as shown in Fig.10.

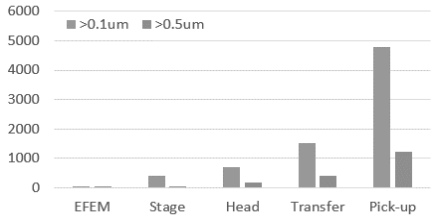

Fig.10 the number of particles generated by area in C2W Bonder

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Thus, a particle suction at the pick-up area, and a particle blowing at the bonding area could be able to lower yield losses induced by particles. However, remounting process, which can fundamentally remove particles generated form diced wafers was much more efficient for stack void reduction. but it cannot but agree that innovative elimination method should be developed.

All-in-one stacking equipment – It takes a few hours to manufacture 1 wafer of 8 high stack. However, as the clean room classification for assembly sites is much more lenient compared to fabrication sites, stacking poses a risk of contamination when the wafers are processed for a long time. We have set-up and used all-in-one stacking equipment to handle pre-treatment process and stacking process altogether. This minimizes particle induced stack voids originating from transportation and storage, as the cleaning process is applied immediately before HBI bonding. In addition, this reduces the cycle time, and eases lot management.

E. Reliability Test Results and Thermal and Electrical Performance of HBI

HBM must withstand complex follow-up processes to be a part of SIP, and thus is inevitably subjected to more mechanical stress than conventional packages. To validate the structural robustness of HBI, a pre-condition test that surpasses moisture sensitivity level 2, and a unbiased-HAST test has been conducted to verify robustness of HBI. The results showed no additional delamination even for voids of 1mm in diameter in HBM cubes with oxide bonding strength of $2J/m^2$.

However, in case of proxy package for system in package, HBI was subjected to more stress due to the greater difference in Coefficient Thermal Expansion (CTE) at the side mold area, than compared to packages with underfills, as shown in Fig.11. Low CTE Epoxy Molding Compound (EMC) was able to release stresses at the triple point between EMC and chips, so it was found that delamination did not occur.

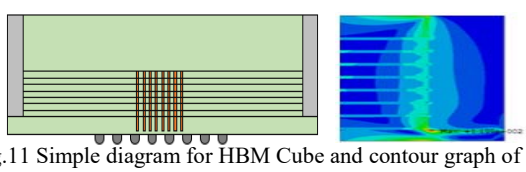

Fig.11 Simple diagram for HBM Cube and contour graph of stress

The HBI structure reduces the interconnection length by more than 100um compared to that of the 8 high micro bump structure, and reduces parasitic interconnection resistance by 19%, inductance by 35% and capacitance by 15%. Thus, the HBI structure can reduce the channel loss from of LC/RC delay in the interconnection. However, the structure of HBM2E exhibits no difference in electrical performance when compared to existing micro bump structures, in 176 evaluation items shown in the table 1.

Table. 1 Electrical performance comparison between HBI and MR-MUF

	Items	Number of Evaluation Items	HBI	MR-MUF
CHDV	IDD0~8	13	22.3W	22.1W
	tCK	4	4.17Gbps	4.17Gbps
	MRS	3	Spec in	Spec in
	Async	21	Spec in	Spec in
	PDN	17	Spec in	Spec in
IFDV	Input Shmoo	10	Spec in	Spec in
	Output Shmoo	3	Spec in	Spec in
FUDV	HBM1	17	> 4.0Gbps	> 4.0Gbps
	HBM2 Gen1	5	> 4.0Gbps	> 4.0Gbps
	HBM2 Gen2	16	> 4.0Gbps	> 4.0Gbps
	HBM2E Func.	63	> 4.0Gbps	> 4.0Gbps
Total		176		

Fig.12 shows that HBM cube with HBI has lower thermal resistance than the current HBM with micro bumps and molded underfills. Even if under-fill materials with high thermal conductivity (~0.7W/℃) and bumps of high density are used, HBM with HBI still outperforms current HBM. Fig.12 shows data that when the bump densities are the same, the HBI structure has a 22.8% lower thermal resistance

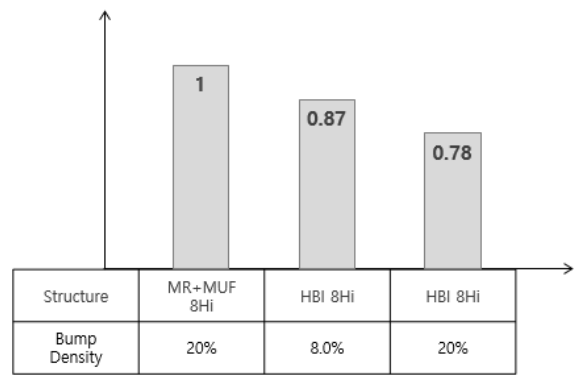

Fig.12 Thermal Resistance Comparison according to each structure (Molded Underfill vs. HBI)

III. CONCLUSIONS

In this paper, we demonstrated 8Hi-HBI packages successfully using the latest real HBM. To obtain robust Cu/Cu joints at low annealing temperatures, special glues for the TBDB process were developed. And CO_2 content were controlled to manage the level of Cu recess between CMP and stacking process aside from precise parameters control of CMP. Also, plasma conditions were optimized for stronger passivation bonds and longer Q-time. As a single void can cause electrical failure and reliability issues, many corrective actions including stacking parameter and particle reduction have been applied to lower stack void. And Low CTE EMC was essential along with strong bonding strength of passivation layers for robust HBI package Finally, HBI has been proven to have no problems in electrical characteristics in real devices manufactured at low annealing temperatures. HBM with HBI

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has a lower thermal resistance than that stacked with micro bumps and underfills.

Although much of the stack voids were removed, a yield loss of over 10% was unavoidable due to foreign materials. We expect new problems to arise in 12 or 16 layer-high stacked HBM, but as many integrated device manufacturing companies, equipment suppliers and material manufacturers interested in multi-stack HBI, we believe that multi-stack HBM with HBI will be mass-produced in the near future.

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